PAPER PRESENTATION

# Programming Refusal with Conditional Activation Steering

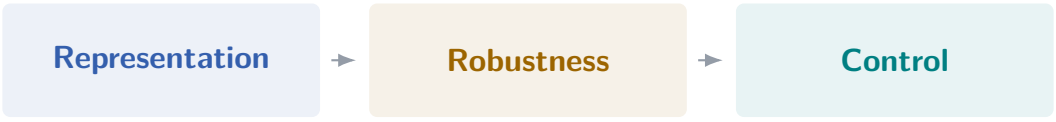
Conditional control of model behavior

Tia Manoukian, Nada Hammad, Nikola Dobričić

Paper by Bruce W. Lee et al. | ICLR 2025

# Does finding a direction mean we can control behavior?

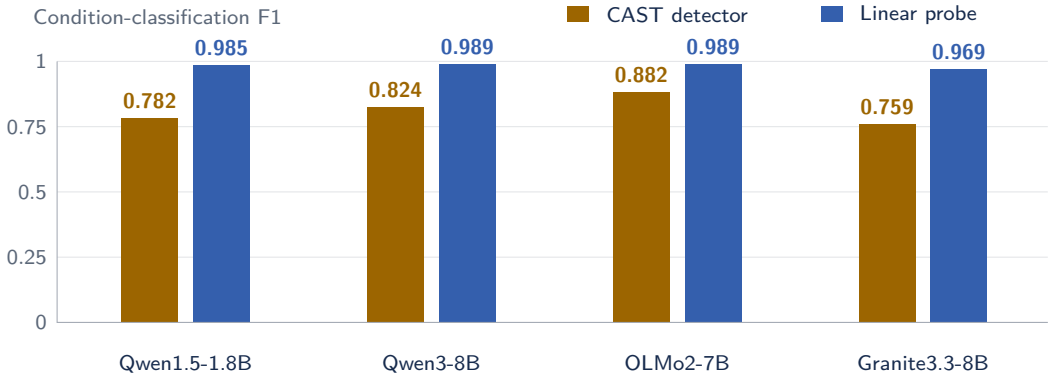
OUR EXPERIMENTS / THE QUESTION




Lee et al., ICLR 2025, §3: condition detection and behavior steering. Our analysis separates the steps.

# The information is there — but CAST doesn't extract all of it

**OUR EXPERIMENTS / REPRESENTATION**




Our controlled ablations | Exp. 5 Held-out F1 · Same hidden-state features

# A harmless change in context can break the detector

OUR EXPERIMENTS / ROBUSTNESS

Original request ⟶ Original request + irrelevant benign text

**Qwen1.5-1.8B**

Harmful prompts

64.6% → 0%

Original + long context

**Qwen3-8B**

Benign prompts

25% → 100%

Original + long context

**Opposite failures. Same loss of selectivity.**

Our controlled ablations | Exp. 4 Gate activation · 16 families per class · 3-seed means

# Detecting a condition is not the same as steering behavior

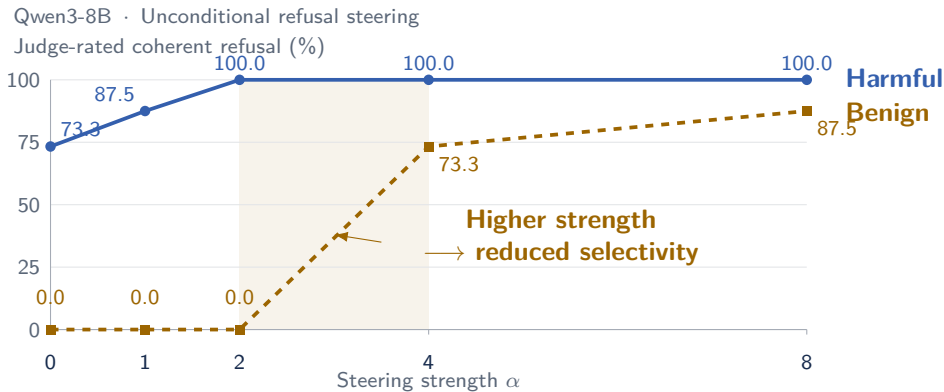
OUR EXPERIMENTS / CONTROL

| Qwen1.5-1.8B | OLMo2-7B |
| --- | --- |
| 68.5% Harmful prompts detected | 82.2% Harmful prompts detected |
| 95.6% Coherent refusal when triggered | 6.5% Coherent refusal when triggered |

Our controlled ablations | Exp. 3 Coherent refusal: model-judge ratings on valid gated harmful responses

# More steering is not necessarily better

**OUR EXPERIMENTS / CONTROL / STRENGTH**




Our controlled ablations | Strength controls 16 families per class · 3-seed means · Valid judgments only

# Three different problems — not one

OUR EXPERIMENTS / THE LESSON

1. **Information in activations**
   Linear probes classify the labels well.
2. **Robust concept detection**
   Context changes can break the gate.
3. **Reliable behavioral control**
   Steering depends on strength and model.

**A decodable representation is not necessarily a robust detector or an effective intervention.**

Our controlled ablations | Synthesis of our controlled ablations  Experiments 3–5 and strength controls

# Does CAST need its nonlinear similarity function?



**CAST similarity**

$$\cos\Big(h, \tanh\big(\mathrm{proj}_c\, h\big)\Big)$$

**0.619**

Held-out F1

**Direct cosine**

$$\cos(h, c)$$

**0.736**

Held-out F1

Qwen1.5-1.8B · Float64 evaluation · Fixed condition layer

Our controlled ablations | Exp. 1 Similarity formula: Lee et al., ICLR 2025, §3.1